1, 12, 15 | 14, 30, 31 | 21, 23, 28 | 36, 38, 41 | 53, 64 | 65, 67 | 77, 79, 86 | 94, 103

1) $\int_{\frac{\pi}{2}}^{\pi} \frac{\sin 2x}{2\sin x}\,dx = ?$

$\sin 2x = 2\sin x\cos x \qquad ? = \int_{\frac{\pi}{2}}^{\pi} \cos x\,dx = \sin x\Big|_{\frac{\pi}{2}}^{\pi} = \sin\pi - \sin\frac{\pi}{2} = 0 - 1 = -1$

2) $\int_{1}^{\sqrt{2}} \frac{x^2 + \sqrt{x}}{x^2}\,dx = ?$

$= \int_1^{\sqrt{2}} 1\,dx + \int_1^{\sqrt{2}} \frac{x^{\frac{1}{2}}}{x^2}\,dx = x\Big|_1^{\sqrt{2}} + \frac{-2}{\sqrt{x}}\Big|_1^{\sqrt{2}} = \sqrt{2} - 1 + \left(\left(-\sqrt{2}\right) - (-2)\right)$

$= \sqrt{2} + 1 - 2^{\frac{3}{4}} = \sqrt{2} + 1 - \sqrt[4]{8}$

$\frac{x^{\frac{1}{2}}}{x^2} = x^{-\frac{3}{2}}$

$\int x^{-\frac{3}{2}}\,dx = -2x^{-\frac{1}{2}} = -\frac{2}{\sqrt{x}}$

3) $\int_{-2}^{0} \frac{x^2 + 5x - 1}{(x-1)(x^2+2x+2)}\,dx = ?$

$\frac{x^2+5x-1}{(x-1)(x^2+2x+2)} = \frac{A}{(x-1)} + \frac{Bx+C}{(x^2+2x+2)} \rightarrow Ax^2 + A2x + 2A + Bx^2 - Bx + Cx - C$

$(x^2+2x+2)\quad (x-1)$

$x^2 + 5x - 1 = Ax^2 + Bx^2 + A2x - Bx + Cx + 2A - C$

$A + B = 1$
$2A - B + C = 5$
$2A - C = -1$

$5A = 5 \rightarrow A = 1,\ B = 0,\ C = 3$
$4A - B = 4$

$? = \int_{-2}^{0} \left(\frac{1}{(x-1)} + \frac{3}{(x^2+2x+2)}\right)dx = (\ln|x-1|)\Big|_{-2}^{0} + 3\int_{-2}^{0} \frac{1}{x^2+2x+2}\,dx$

$$\int_{-2}^{0} \frac{1}{x^2+2x+2}\,dx = \int_{-2}^{0} \frac{1}{(x+1)^2+1} = \arctan(x+1)$$

$$?= \ln|-1| - \ln|-3| + 3\left[\left(\arctan(1) - \arctan(-1)\right)\right]$$
$$= -\ln 3 + 3\left[\left(\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right)\right] = -\ln 3 + \frac{3\pi}{2}$$

N4) $\int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin^2 x}\,dx = ?$

$\sin^2 x + \cos^2 x = 1$

$\sin 2x = 2\sin x \cos x$

$\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x = \cos^2 x - \sin^2 x$

$\sin^2 x = \frac{1-\cos 2x}{2} \qquad \cos^2 x = \frac{\cos 2x + 1}{2}$

anlk

$$?= \int_0^{\frac{\pi}{2}} \frac{\cos x}{1+\sin^2 x}\,dx \Rightarrow \int \frac{dv}{1+v^2} = \arctan(v) = \arctan(\sin x) \Rightarrow \arctan(\sin x)\Big|_0^{\frac{\pi}{2}}$$

$$= \arctan\left(\sin\tfrac{\pi}{2}\right) - \arctan(\sin 0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$\sin x = v$

$\cos x\,dx = dv$

S1) $\int_1^x 2f'(t)f(t)\,dt = (x^2-1)^2$ ve $f(1)=0$ ise $f(x)=?$

$f(t) = v$

$f'(t)\,dt = dv$

$\rightarrow 2\int v\,dv \rightarrow 2\frac{v^2}{2} \rightarrow f^2(t)\Big|_1^x = f^2(x) - f^2(1) = (x^2-1)^2$

$f^2(x) = (x^2-1)^2$

$f(x) = |x^2-1|$

6) a) $\int_0^1 \frac{x^3}{x^8+1}\,dx = ?$

$$\int_0^1 \frac{x^3}{(x^4)^2+1}\,dx \to \begin{pmatrix} x^4 = v \\ 4x^3\,dx = dv \end{pmatrix} \to \frac{1}{4}\int \frac{dv}{v^2+1} = \frac{\arctan(v)}{4} = \frac{\arctan(x^4)}{4}$$

$$= \frac{\arctan(x^4)}{4}\Big|_0^1 = \frac{\arctan(1)}{4} - \frac{\arctan(0)}{4} = \frac{\pi}{16}$$

b) $\int_1^e \frac{\sin(\ln x)}{x}\,dx = ?$

$$\begin{pmatrix} \ln x = v \\ \frac{1}{x}dx = dv \end{pmatrix} \to \int \sin(v)\,dv = -\cos(v) \to -\cos(\ln x)\Big|_1^e = -\cos(\ln e) + \cos(\ln 1)$$
$$-\cos 1 + 1$$

c) $\int_{-1}^1 \frac{3x+8}{x^2+2x}\,dx = ?$

$$\frac{3x+8}{x^2+2x} = \frac{A}{x} + \frac{B}{x+2} \to 3x+8 = Ax+2A+Bx$$

$$A+B=3$$
$$2A=8 \to A=4,\ B=-1$$

$$\int_{-1}^1 \left(\frac{4}{x} + \frac{-1}{x+2}\right)dx = (4\ln|x| - \ln|x+2|)\Big|_{-1}^1 = \cancel{4\ln 1} - \ln 3 - \cancel{4\ln 1} + \ln|1| = -\ln 3$$

7) a) $\int_0^1 \frac{x^3}{x+2}\,dx = ?$

$$\begin{array}{r|l} x^3 & x+2 \\ -x^3-2x^2 & x^2-2x+4 \\ \hline -2x^2 & \\ -2x^2-4x & \\ \hline 4x+8 & \\ -4x-8 & \\ \hline 0 & \end{array}$$

$$?= \int_0^1 \frac{(x+2)(x^2-2x+4)-8}{(x+2)}\,dx = \int_0^1 \left((x^2-2x+4) - \frac{8}{x+2}\right)dx$$

$$= \left(\frac{x^3}{3} - x^2 + 4x\right)\Big|_0^1 - 8\ln|x+2|\Big|_0^1 = \frac{1}{3} - 1 + 4 - 8(\ln|3| - \ln|2|)$$

$$= \frac{10}{3} - 8\ln\left|\frac{3}{2}\right|$$

d) $\int \frac{\ln(\sin x)}{\sin^2 x}\,dx = ?$

$\left(\frac{1}{\sin^2 x} = \operatorname{cosec}^2 x \quad \int \operatorname{cosec}^2 x = -\cot x + c\right)$

$\left(\text{Laplace} \quad \ln(\sin x) = u \quad \frac{dx}{\sin^2 x} = dv \quad \int \operatorname{cosec}^2 x\,dx = \int dv \quad -\cot x = v\right)$

$u \cdot v - \int v \cdot du$

$$\int \frac{\ln(\sin x)}{\sin^2 x}\,dx = \ln(\sin x)\cdot(-\cot x) + \int \cot^2 x\,dx = -\cot x \cdot \ln(\sin x) - \cot x - x + c$$

$$\int (\csc^2 x - 1)\,dx = -\cot x - x + c$$

$(\cot^2 x = \operatorname{cosec}^2 x - 1)$

g) $\int_0^{4\pi} \sqrt{1-\cos 2x}\,dx = ?$

$(\cos 2x = 2\cos^2 x - 1 = 1 - 2\sin^2 x) \rightarrow \int_0^{4\pi} \sqrt{2\sin^2 x}\,dx = \int_0^{4\pi} \sqrt{2}\cdot|\sin x|\,dx$

$$\sqrt{2}\int_0^{4\pi} |\sin x|\,dx = 4\sqrt{2}\int_0^{\pi} |\sin x|\,dx = 4\sqrt{2}(-\cos x)\Big|_0^{\pi} = 4\sqrt{2}(-\cos\pi - (-\cos 0))$$

$$= 8\sqrt{2}$$

$[0,\pi]$ aralığında $\sin x \geq 0$

$[\pi, 2\pi]$ " $\sin x \leq 0 \rightarrow |\sin x| = -\sin x$ (tamsa alan aynı)

14) $\int_0^{2\pi} \left|x - \frac{\pi}{2}\right| \sin x \, dx = ?$

kritik $\frac{\pi}{2}$

$$? = \int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right) \sin x \, dx + \int_{\frac{\pi}{2}}^{2\pi} \left(x - \frac{\pi}{2}\right) \sin x \, dx$$

! $\int (x-k) \sin x \, dx$ → $(x-k)(-\cos x) + \sin x + C$

Laplu

$\begin{pmatrix} x-k = u & \sin x dx = dv \\ dx = du & -\cos x = v \end{pmatrix}$ $u \cdot v - \int v \cdot du$

$$? = \left(\left(x - \frac{\pi}{2}\right)(\cos x) - \sin x\right)\Big|_0^{\frac{\pi}{2}} + \left(\left(x - \frac{\pi}{2}\right)(-\cos x) + \sin x\right)\Big|_{\frac{\pi}{2}}^{2\pi}$$

$$= \left(0 - \sin\frac{\pi}{2}\right) - \left(\left(-\frac{\pi}{2}\right)\cos 0 - \sin 0\right) + \left(\left(\frac{3\pi}{2}\right)(-\cos 2\pi) + \sin 2\pi\right) - \left(0 + \sin\frac{\pi}{2}\right)$$

$$= -1 - \left(\left(-\frac{\pi}{2}\right) - 0\right) + \left(-\frac{3\pi}{2} + 0\right) - (0 + 1)$$

$$= -1 + \frac{\pi}{2} - \frac{3\pi}{2} - 1 = -2 - \pi$$

15) a) $\int_{-2}^{1} (\llbracket x \rrbracket + |x|) \, dx = ?$

$-2 \leq x < -1 \rightarrow \llbracket x \rrbracket = -2$

$-1 \leq x < 0 \rightarrow \llbracket x \rrbracket = -1$

$0 \leq x < 1 \rightarrow \llbracket x \rrbracket = 0$

$$? = \int_{-2}^{-1} (-2 - x) \, dx + \int_{-1}^{0} (-1 - x) \, dx + \int_{0}^{1} (0 + x) \, dx$$

$$= \left(-2x - \frac{x^2}{2}\right)\Big|_{-2}^{-1} + \left(-x - \frac{x^2}{2}\right)\Big|_{-1}^{0} + \left(\frac{x^2}{2}\right)\Big|_{0}^{1}$$

$= \left(2 - \frac{1}{2}\right) - (+4 - 2) + (0) - \left(+1 - \frac{1}{2}\right) + \left(\frac{1}{2}\right) - 0$

$= \left(\frac{3}{2} - 2\right) - \frac{1}{2} + \frac{1}{2} = \underline{\underline{-\frac{1}{2}}}$

b) $\int_{-4}^{4} |(2t^2 + 3|t| + 5)|\,dt = ?$

↳

! - Bu terimlerin toplamı daima pozitiftir. Dıştaki mutlak etkisizdir.
- $f(t) = f(-t)$ old. çift fonksiyondur yani y'ye göre simetriktir.

$? = 2 \cdot \int_{0}^{4} (2t^2 + 3|t| + 5)\,dt = 2 \cdot \int_{0}^{4} (2t^2 + 3t + 5)\,dt = \underline{\underline{\frac{520}{3}}}$

4) $\int_{-1}^{2} |x-1|\,(\operatorname{sgn} x)\,dx = ?$

$|x-1| = \begin{cases} 1-x, & x \le 1 \\ x-1, & x > 1 \end{cases}$ $\quad \operatorname{sgn} x = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$

$? = \int_{-1}^{0} (1-x)\cdot(-1)\,dx + \int_{0}^{1} (1-x)(1)\,dx + \int_{1}^{2} (x-1)(1)\,dx$

$= \left(\frac{x^2}{2} - x\right)\Big|_{-1}^{0} + \left(x - \frac{x^2}{2}\right)\Big|_{0}^{1} + \left(\frac{x^2}{2} - x\right)\Big|_{1}^{2}$

$= 0 - \left(\frac{1}{2} + 1\right) + \left(1 - \frac{1}{2}\right) - 0 + (2 - 2) - \left(\frac{1}{2} - 1\right)$

$= -\frac{3}{2} + \frac{1}{2} + \frac{1}{2} = \underline{\underline{-\frac{1}{2}}}$

3) Eğer f, a'yı kapsayan bir açık aralıkta sürekli olduğuna göre

$\lim_{x \to a} \dfrac{\int_{a}^{x} \sin(f(t))\,dt}{\int_{a}^{x} f(t)\,dt}$ limitini hesaplayınız.

$$\lim_{x\to a}\frac{x\cdot\int_a^x \sin(f(t))\,dt}{\int_a^x f(t)\,dt}=\frac{a\cdot 0}{0}=\frac{0}{0}\ \text{bls.}$$

Kural: $\frac{d}{dx}\left(\int_{g(x)}^{f(x)} h(t)\,dt\right)=h(f(x))\cdot f'(x)-h(g(x))\cdot g'(x)$

- $\frac{d}{dx}\left(\int_a^x \sin(f(t))\,dt\right)=\sin(f(x))\cdot 1-0$
- $\frac{d}{dx}\left(\int_a^x f(t)\,dt\right)=f(x)\cdot 1-0$

$$\frac{d}{dx}\left(x\cdot\int_a^x \sin(f(t))\,dt\right)=1\int_a^x \sin(f(x))\,dt+x\cdot\sin(f(x))$$

$$?=\lim_{x\to a}\frac{\int_a^x \sin(f(x))\,dt+x\cdot\sin(f(x))}{f(x)}\overset{x=a}{=}\frac{0+a\cdot\sin(f(a))}{f(a)}=\frac{a\cdot\sin(f(a))}{f(a)}$$

$$\lim_{x\to a}\frac{\int_a^x \sin(f(x))\,dt+x\cdot\sin(f(x))}{f(x)}=\frac{a\cdot\sin(0)}{0}=\frac{0}{0}$$

$f(a)=0$ ise yine $\frac{0}{0}$ bls.

$f(a)\neq 0$ ise sonuç bu

$$\lim_{x\to a} a\cdot\frac{\sin(f(x))}{f(x)}\to a\cdot\lim_{u\to 0}\frac{\sin(u)}{u}=a\cdot 1=a \quad \text{sonuç}$$

$\left(\begin{matrix}u=f(x)\\ du=f'(x)dx\end{matrix}\right)$ $\boxed{\lim_{u\to 0}\frac{\sin u}{u}=1}$

$$\begin{cases}\frac{a\cdot\sin(f(a))}{f(a)}, & f(a)\neq 0\\ a, & f(a)=0\end{cases}$$

20) $\lim_{x\to 0} \frac{\int_0^{x^2} \sin t^2\,dt}{2x^6} = ?$

$= \frac{0}{0}$ bls. $\xrightarrow{\text{L'hosp.}} \lim_{x\to 0} \frac{\sin 0 \cdot 0 - (\sin x^4)\,2x}{12x^5} = \frac{0}{0}$ bls

$\xrightarrow{\text{L'hosp.}} -\lim_{x\to 0} \frac{8x^4 \cdot \cos x^4 + 2\cdot \sin x^4}{60x^4} = -\left(\lim_{x\to 0} \frac{2\cos x^4}{15} + \lim_{x\to 0} \frac{2\sin x^4}{60x^4}\right)$

$\frac{2}{15}$ $\quad /L'H$

$= -\frac{5}{30} = -\frac{1}{6}$

$\lim_{x\to 0} \frac{8x^3 \cdot \cos x}{60 \cdot 4 \cdot x^3} = \frac{1}{30}$

---

21) $\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n}\right) = ?$

$= \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{n+k} = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \frac{1}{1+\frac{k}{n}}$

$f\left(\frac{k}{n}\right) = \frac{1}{1+\frac{k}{n}} \xrightarrow{\frac{k}{n}=x} f(x) = \frac{1}{1+x}$

$\sum_{k=1}^{n} \frac{1}{n\left(1+\frac{k}{n}\right)} = \frac{1}{n} \sum_{k=1}^{n} \frac{1}{1+\frac{k}{n}}$

$\int_0^1 f(x)\,dx = \int_0^1 \frac{1}{1+x}\,dx = \ln|1+x|_0^1$

$= \ln 2 - \ln 1 = \ln 2$

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23) $\lim_{n\to\infty} \left(\frac{n}{n^2+1} + \frac{n}{n^2+4} + \cdots + \frac{n}{n^2+n^2}\right) = ?$

$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{n}{n^2+k^2} = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \frac{n}{n+\frac{k^2}{n}}$

$f\left(\frac{k}{n}\right) = \frac{n}{n+\frac{k^2}{n}} = \frac{n}{n\left(1+\frac{k^2}{n^2}\right)} = \frac{1}{1+\left(\frac{k}{n}\right)^2}$

$f(x) = \frac{1}{1+x^2}$

$\sum_{k=1}^{n} \frac{n}{n\left(n+\frac{k^2}{n}\right)} = \frac{1}{n} \sum_{k=1}^{n} \frac{n}{n+\frac{k^2}{n}}$

$$\int_0^1 f(x)dx = \boxed{\int_0^1 \frac{1}{1+x^2}dx = (\arctan x)\Big|_0^1} = \arctan 1 - \arctan 0 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

~~281~~ $\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n} = ?$

$\lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{\frac{1}{n}} = 0^0$ bls.

$y = \left(\frac{n!}{n^n}\right)^{\frac{1}{n}} \xrightarrow{\ln} \ln y = \ln\left(\frac{n!}{n^n}\right)^{\frac{1}{n}} \Rightarrow \ln y = \frac{1}{n}\ln\left(\frac{n!}{n^n}\right) \xrightarrow{\lim} \lim_{n\to\infty}\ln y = \lim_{n\to\infty}\frac{1}{n}\cdot\ln\left(\frac{n!}{n^n}\right)$

$= 0 \cdot -\infty$

$\ln a.b = \ln a + \ln b$

$$\lim_{n\to\infty}\frac{1}{n}\cdot\ln\left(\frac{n!}{n^n}\right) = \lim_{n\to\infty}\frac{1}{n}\left(\ln\frac{1}{n} + \ln\frac{2}{n} + \dots \ln\frac{n}{n}\right) = \lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\ln\left(\frac{k}{n}\right)$$

$f\left(\frac{k}{n}\right) = \ln\left(\frac{k}{n}\right) \xrightarrow{\frac{k}{n}=x} f(x) = \ln x \qquad \int_0^1 f(x)dx = \int_0^1 \ln x\,dx = (x\ln x - x)\Big|_0^1$

$1\cdot\ln 1 - 1 - (0\cdot\ln 0 - 0) = -1$

$0 \cdot -\infty$

$\lim_{x\to 0} x\cdot\ln x = \frac{\ln x}{\frac{1}{x}} = \frac{-\infty}{\infty}$ bls. L.H.

$\lim_{x\to 0}\frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x\to 0} -x = 0$

$\lim_{n\to\infty}\ln y = -1 \qquad \lim_{n\to\infty} y = e^{-1}$

36) $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$ mt. yakınsaklık ?

$$? = \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_{0}^{\infty} \frac{dx}{1+x^2} = \lim_{t\to-\infty} \int_{t}^{0} \frac{dx}{1+x^2} + \lim_{t\to\infty} \int_{0}^{t} \frac{dx}{1+x^2}$$

$$\lim_{t\to-\infty} (\arctan x)\Big|_{t}^{0} + \lim_{t\to\infty} (\arctan x)\Big|_{0}^{t} = \arctan 0 - \arctan t + \arctan t - \arctan 0$$

$$= -\left(-\frac{\pi}{2}\right) + \frac{\pi}{2} = \pi \text{ (yakınsak)}$$

37) $\int_{0}^{\infty} \frac{dx}{x-1}$ mt yakınsaklık ?

$$? = \lim_{t\to\infty} \int_{0}^{t} \frac{dx}{x-1} = \lim_{t\to\infty} (\ln|x-1|)\Big|_{0}^{t} = \lim_{t\to\infty} \ln|t-1| - \ln 1 = \ln\infty = \infty$$

(ıraksak)

44) $f(t) = \int_{0}^{\infty} x^{t-1} e^{-x} dx$, $f\left(\frac{1}{2}\right) = \sqrt{\pi}$, $f(n+1) = nf(n) = n!$

$$B(m,n) = \int_{0}^{1} x^{m-1}(1-x)^{n-1} dx, \quad B(m,n) = \frac{f(m).f(n)}{f(m+n)}$$

a) $\int_{0}^{\infty} x^6 . e^{-2x} dx = ?$ ($2x = v$ dön. yap)

$$\left(\begin{matrix} 2x = v \\ 2dx = dv \end{matrix}\right) \quad \frac{1}{128} \int_{0}^{\infty} v^6 . e^{-v} dv = \frac{1}{128} f(7) = \frac{720}{128} = \frac{45}{8}$$

$f(t)$

$$f(7) = f(6+1) = 6.f(6) = 6! = 720$$

b) $\int_{0}^{2} \frac{x^2}{\sqrt{2-x}} dx = ?$ ($x = 2v$ dön. yap)

$x = 2v \quad dx = 2dv \qquad x=2 \to v=1 \quad x=0 \to v=0$

$$? = \int_0^1 \frac{4v^2 \cdot 2dv}{\sqrt{2-2v}} = \frac{8}{\sqrt{2}} \int_0^1 \frac{v^2 dv}{\sqrt{1-v}} = \frac{8}{\sqrt{2}} \int_0^1 v^2 (1-v)^{-\frac{1}{2}} dv$$

$$= \frac{8}{\sqrt{2}} \cdot B\left(3, \tfrac{1}{2}\right)$$

$$= 4\sqrt{2} \cdot B\left(3, \tfrac{1}{2}\right) = 4\sqrt{2} \cdot \left(\frac{\Gamma(3)\cdot\Gamma(\frac{1}{2})}{\Gamma(3+\frac{1}{2})}\right) = 4\sqrt{2}\left(\frac{2\cdot\sqrt{\pi}}{\frac{15}{8}\sqrt{\pi}}\right) = \frac{64\sqrt{2}}{15}$$

ek) $\int_0^2 x\sqrt[3]{8-x^3}\,dx = ?$ ($x^3 = 8y$ dön. kullan.)

$x^3 = 8y \qquad x = 2y^{1/3} \qquad dx = \frac{2}{3} y^{-2/3} dy$

$x=2 \to y=1 \qquad x=0 \to y=0$

$$? = \int_0^1 2y^{1/3} \sqrt[3]{8-8y}\, \frac{2}{3} y^{-\frac{2}{3}} dy$$

$$= 2 \cdot 2 \cdot \frac{2}{3} \int_0^1 y^{-1/3} (1-y)^{1/3} dy$$

$$= B\left(\tfrac{2}{3}, \tfrac{4}{3}\right)$$

$$? = \frac{8}{3} \cdot B\left(\tfrac{2}{3}, \tfrac{4}{3}\right) = \frac{16\pi}{9\sqrt{3}}$$

---

53) $y^2 = x$, $y = 2x - 1$ eğrileri arasında kalan bölge alanı?

$x = y^2 \qquad y = 2x - 1$

$\frac{y+1}{2} = x$

$y^2 = \frac{y+1}{2}$

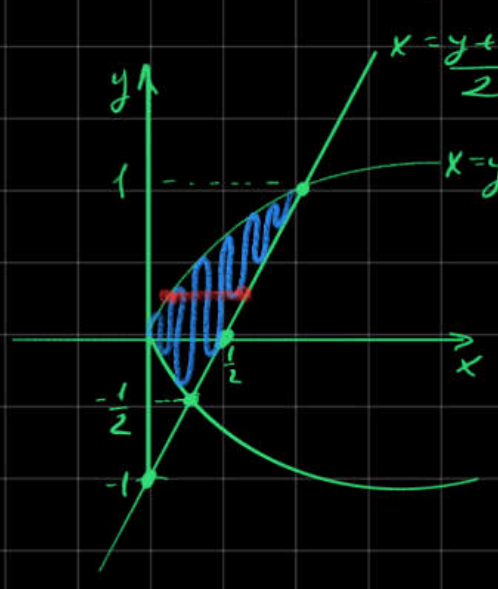
$2y^2 - y - 1 = 0$

$y \quad -1$

$2y \quad 1$

$(y-1)(2y+1) = 0$

$y = 1, \; y = -\frac{1}{2}$



$$A = \int_{-\frac{1}{2}}^{1} \left(\left(\frac{y+1}{2}\right) - y^2\right) dy$$

$$= \int_{-\frac{1}{2}}^{1} \left(\frac{y}{2} + \frac{1}{2} - y^2\right) dy = \left(\frac{y^2}{4} + \frac{y}{2} - \frac{y^3}{3}\right)\Big|_{-\frac{1}{2}}^{1} = \frac{27}{48}$$

(64) $x = a(t - \sin t)$ ve $y = a(1 - \cos t)$ denklemleri ile verilen sikloidin birinci yayı ile x-ekseni arasında kalan alanı bulunuz.

$y = 0 = a(1 - \cos t)$

$\cos t = 1$

$t = 0 \quad \vee \quad t = 2\pi$

(yayın başlangıç ve bitiş noktası)

$x = a(t - \sin t)$

$\frac{dx}{dt} = a(1 - \cos t)$

$dx = a(1 - \cos t)\,dt$

$$A = \int_0^{2\pi} a(1-\cos t).\, a(1-\cos t)\,dt = a^2 \int_0^{2\pi} (1-\cos t)^2\,dt$$

$$= a^2 \int_0^{2\pi} (1 - 2\cos t + \cos^2 t)\,dt = a^2 \int_0^{2\pi} \left(\frac{3}{2} - 2\cos t + \frac{\cos(2t)}{2}\right) dt$$

$\cos^2 t = \frac{1 + \cos(2t)}{2}$

$$= a^2 \cdot \left(\frac{3t}{2} - 2\sin t + \frac{\sin 2t}{2 \cdot 2}\right)\Big|_0^{2\pi} = 3\pi \cdot a^2$$

$\int \sin x\,dx = -\cos x$

$\int \cos x\,dx = \sin x$

---

(65) $r = \frac{4}{2\cos\theta - \sin\theta}$ eğrisinin kartezyen karşılığı?

$x = r\cos\theta \longrightarrow \cos\theta = \frac{x}{r} = \frac{x}{\sqrt{x^2+y^2}}$

$y = r\sin\theta \longrightarrow \sin\theta = \frac{y}{r} = \frac{y}{\sqrt{x^2+y^2}}$

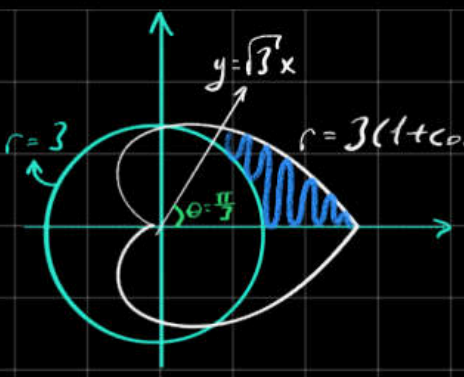
$r^2 = x^2 + y^2 \longrightarrow r = \sqrt{x^2+y^2}$

$$\sqrt{x^2+y^2} = \frac{4}{2x-y} = 4r = 4\sqrt{x^2+y^2} \longrightarrow 1 = \frac{4}{2x-y}$$

$2x - y = 4$

$y - 2x + 4 = 0$

67) (a)



taralı alan?

$$A = \frac{1}{2}\int_{\theta_1}^{\theta_2} r^2\,d\theta$$

$y = \sqrt{3}\cdot x \quad / \sin\theta = \sqrt{3}\cdot/\cos\theta \quad | \quad \tan\theta = \sqrt{3}$

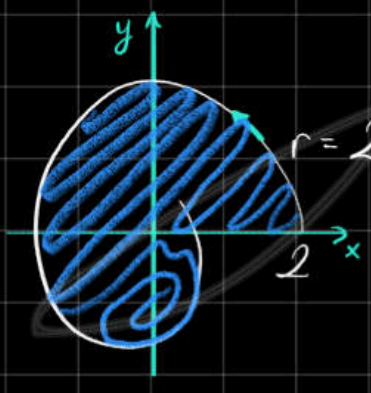
$0 = \sqrt{3}\cos\theta - \sin\theta \quad | \quad 60^\circ = \frac{\pi}{3}$

$$\frac{1}{2}\int_0^{\frac{\pi}{3}}\left((3(1+\cos\theta))^2 - (3)^2\right)d\theta = \frac{1}{2}\int_0^{\frac{\pi}{3}}(9+18\cos\theta+9\cos^2\theta-9)\,d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{3}}(18\cos\theta+9\cos^2\theta)\,d\theta = \frac{9}{2}\int_0^{\frac{\pi}{3}}\left(2\cos\theta+\frac{1+\cos 2\theta}{2}\right)d\theta$$

$$= \frac{9}{2}\left(2\sin\theta+\frac{\theta}{2}+\frac{\sin 2\theta}{4}\right)\Big|_0^{\frac{\pi}{3}} = \frac{81\sqrt{3}}{16}+\frac{3\pi}{4}$$

(b)



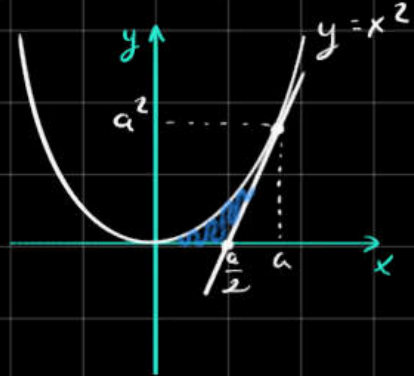
taralı alan?

$$\int e^{2x}\,dx = \frac{e^{2x}}{2}+c$$

$$A = \frac{1}{2}\int_0^{2\pi}(2\cdot e^{-\frac{\theta}{10}})^2\,d\theta = \frac{1}{2}\int_0^{2\pi}4\cdot e^{-\frac{\theta}{5}}\,d\theta = 2\cdot\left(\frac{1}{-\frac{1}{5}}\cdot e^{-\frac{\theta}{5}}\right)\Big|_0^{2\pi} = -10e^{-\frac{2\pi}{5}}+10$$

---

77)



B bölgeyi X ve Y ekseni etrafında dönd. olusan cismin hacmi?

$y' = 2x \quad y'(a) = 2a \rightarrow y - a^2 = 2a(x-a)$

$y = 2ax - a^2$

$$V_1 = \pi \int_0^a ((x^2)^2\,dx = \frac{\pi a^5}{5}$$

$$V_2 = \pi \int_{\frac{a}{2}}^{a} (2ax - a^2)^2\,dx = \frac{\pi}{2a}\int_0^{a^2} u^2\,du = \frac{\pi a^5}{6}$$

$$\left\{ V_1 - V_2 = \frac{\pi a^5}{30} \right. \quad \left(\begin{matrix} x \text{ eks.} \\ \text{etr.} \end{matrix} \right)$$

$$\left(\begin{matrix} 2ax - a^2 = u & x = \frac{a}{2} \to u = 0 \\ 2a\,dx = du & x = a \to u = a^2 \end{matrix} \right)$$

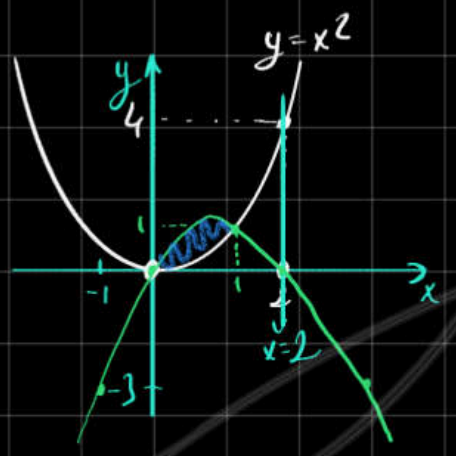
$y = x^2$
$x = \sqrt{y}$

$y = 2ax - a^2$
$x = \frac{y + a^2}{2a}$

$$V = \pi \int_0^{a^2} \left(\left(\frac{y+a^2}{2a} \right)^2 - (\sqrt{y})^2 \right) dy = \frac{\pi a^4}{12}$$

(y eks. etrafında)

7(9) $y = x^2$, $y = 2x - x^2$ paraboller arasında kalan bölge $x = 2$ doğrusu et. dönd. Oluşan dönel cismin hacmini bulunuz.



$$V = 2\pi \int_{x_1}^{x_2} r \cdot h \cdot dx$$

$r = 2 - x$

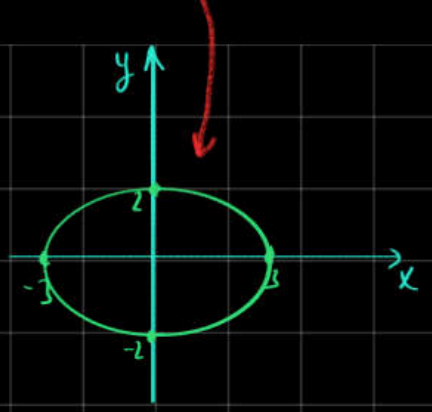
$h = (2x - x^2) - (x^2) = 2x - 2x^2$

$$V = 2\pi \int_0^1 (2 - x)(2x - 2x^2)\,dx = \pi$$

$x^2 = 2x - x^2$
$2x^2 = 2x$
$x = 1$

8(6) $\frac{x^2}{9} + \frac{y^2}{4} = 1$ denklemi elipsin

a) x eks. et. dönd. ol. cismin hacmi?
b) y eks " " " " " ?



$y^2 = 4(1-\frac{x^2}{9})$ $\quad x^2 = 9(1-\frac{y^2}{4})$

x eks. etr. $V_x = \pi \int_{-3}^{3} y^2\,dx = 16\pi$

y eks. etr. $V_y = \pi \int_{-2}^{2} x^2\,dy = 24\pi$

9(4) $x = \frac{1}{4}y^2 - \frac{1}{2}\ln y$ eğrisinin ordinatları $y_1 = 1$ ve $y_2 = 2$ olan noktaları arasında kalan kısmın uzunluğunu bulunuz.

$$\ell = \int_{x_1}^{x_2} \sqrt{1+(f'(x))^2}\,dx \qquad \ell = \int_{y_1}^{y_2} \sqrt{1+(f'(y))^2}\,dy$$

$$f'(y) = \frac{y}{2} - \frac{1}{2y} = \frac{y^2-1}{2y}$$

$$(f'(y))^2 = \frac{y^4-2y^2+1}{4y^2} \qquad (f'(y))^2+1 = \frac{y^4-2y^2+4y^2+1}{4y^2} = \frac{y^4+2y^2+1}{4y^2} = \left(\frac{y^2+1}{2y}\right)^2$$

$$\ell = \int_1^2 \sqrt{\left(\frac{y^2+1}{2y}\right)^2}\,dy = \int_1^2 \left(\frac{y^2+1}{2y}\right)dy = \frac{1}{4}\left(\frac{y^2}{2}+\ln y\right)\Big|_1^2 = \frac{3}{4}+\frac{\ln 2}{2}$$

$$\left(\frac{1}{2}\int \frac{(y^2+1)}{y}\,dy = \frac{1}{4}\int \left(y+\frac{1}{y}\right)dy = \frac{1}{2}\left(\frac{y^2}{2}+\ln y\right)\right)$$

$y^2+1 \;\big|\; y$

$-y^2 \quad\; y$

1

~~103~~ $y = \cos x$ eğrisinin $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ aralığındaki kısmının X-ekseni et. dönd. ile elde edilen dönel cismin alanı?

$$S = 2\pi \int_{x_1}^{x_2} |f(x)| \sqrt{1+(f'(x))^2}\, dx$$

$f(x) = \cos x$

$f'(x) = -\sin x$

$(f'(x))^2 = \sin^2 x$

$(f'(x))^2 + 1 = \sin^2 x + 1$

Sift fonk.

$$S = 2\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} |\cos x| \sqrt{1+\sin^2 x}\, dx$$

$$= 4\pi \int_{0}^{\frac{\pi}{2}} |\cos x| \sqrt{1+\sin^2 x}\, dx$$

$\left(\begin{array}{ll} \sin x = u & x = \frac{\pi}{2} \to u = 1 \\ \cos x\, dx = du & x = 0 \to u = 0 \end{array}\right)$

$$= 4\pi \int_{0}^{1} \sqrt{1+u^2}\, du$$

bu ne la

$$= 4\pi \cdot \frac{1}{2}\left(u\sqrt{1+u^2} + \ln\left|u+\sqrt{1+u^2}\right|\right)\Big|_0^1 = 2\pi\left(\sqrt{2} + \ln(1+\sqrt{2})\right)$$

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41) $\int_{-\pi}^{\pi} \cos(|x|+x)\,dx = ?$

$= \int_{-\pi}^{0} \cos 0\,dx + \int_{0}^{\pi} \cos 2x\,dx = x\Big|_{-\pi}^{0} + \frac{\sin 2x}{2}\Big|_{0}^{\pi} = +\pi + \underbrace{\left(\frac{\sin 2\pi}{2} - \frac{\sin 0}{2}\right)}_{0} = \pi$

28) $\lim_{n\to\infty} \left(\frac{n!}{n^n}\right)^{1/n} = ?$

$y = \left(\frac{n!}{n^n}\right)^{1/n} \xrightarrow{\ln} \ln y = \ln\left(\frac{n!}{n^n}\right)^{1/n} = \frac{1}{n}\cdot \ln\left(\frac{n!}{n^n}\right) \xrightarrow{\lim} \lim_{n\to\infty} \ln y = \lim_{n\to\infty} \frac{1}{n} \ln\left(\frac{n!}{n^n}\right)$

$(\ln(a\cdot b) = \ln a + \ln b)$

$\lim_{n\to\infty} \frac{1}{n} \ln\left(\frac{n!}{n^n}\right) = \lim_{n\to\infty} \frac{1}{n}\left(\frac{\ln 1}{n} + \frac{\ln 2}{n} + \dots \frac{\ln n}{n}\right) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \ln\left(\frac{k}{n}\right)$

$f\left(\frac{k}{n}\right) = \ln\left(\frac{k}{n}\right) \xrightarrow{\frac{k}{n}=x} f(x) = \ln(x) \longrightarrow \int_0^1 f(x)\,dx = \int_0^1 \ln(x)\,dx = (x\ln x - x)\Big|_0^1$

$= \underbrace{(1\ln 1 - 1)}_{-1} - (\underbrace{0\cdot \ln 0}_{-\infty} - 0) = -1 - 0 = -1$

bls. $\lim_{x\to 0} x\cdot \ln x = 0\cdot\infty$ bls. $\to \lim_{x\to 0} \frac{\ln x}{\frac{1}{x}} = \frac{\infty}{\infty}$ bls.

$\lim_{x\to 0} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \lim_{x\to 0} -x = 0$

$\lim_{x\to\infty} \ln y = -1$

$\lim_{x\to\infty} y = e^{-1}$

33) $\lim_{x\to\frac{\pi}{2}} \frac{\frac{ex^2}{\pi} - \frac{e\pi}{4} + \int_x^{\frac{\pi}{2}} e^{\sin t}\,dt}{1+\cos 2x} = ?$

$\lim_{x\to\frac{\pi}{2}} \frac{\frac{ex^2}{\pi} - \frac{e\pi}{4} + \int_x^{\frac{\pi}{2}} e^{\sin t}\,dt}{1+\cos 2x} = \frac{0}{0}$ bls. $\xrightarrow{\text{L.H.}} \frac{\frac{e\cdot 2x}{\pi} + (0 - (e^{\sin x})(1))}{-2\sin 2x} \to \lim_{x\to\frac{\pi}{2}} \frac{\frac{e\cdot 2x}{\pi} - e^{\sin x}}{-2\sin 2x} = \frac{0}{0}$ bls.

L.H.

$$\lim_{x\to\frac{\pi}{2}} \frac{\frac{2e}{\pi} - \cos x \cdot e^{\sin x}}{-4\cos 2x} = \frac{\frac{2e}{\pi} - 0}{4} = \frac{e}{2\pi}$$

7b) $x = \frac{3}{4}$ doğrusunun sağında $r = 1+\cos\theta$ kardioidinin içinde kalan bölgenin alanı?

θ	0	60°	180°	270°	360°
r	2	1	0	1	2



$$A = \frac{1}{2}\int r^2 d\theta$$

$x = \frac{3}{4} \to r\cos\theta = \frac{3}{4}$

$r = \frac{3}{4\cos\theta}$

$$\frac{3}{4\cos\theta} = 1+\cos\theta$$

$$3 = 4\cos\theta + 4\cos^2\theta$$

$$0 = 4x^2 + 4x - 3 = (2\cos\theta + 3)(2\cos\theta - 1)$$

$2x \quad +3$

$2x \quad -1$

$\cos\theta = -\frac{3}{2}$ (✗) $\qquad \cos\theta = \frac{1}{2}$

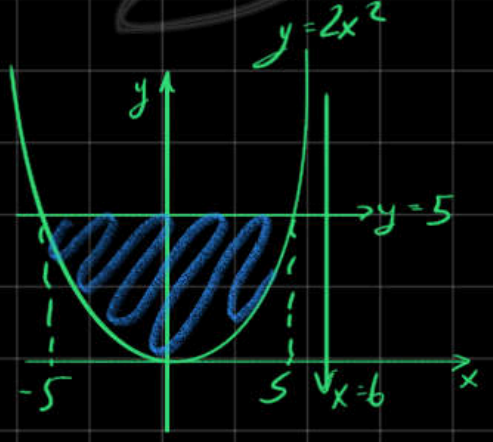
$\pm\frac{\pi}{3}$

$$\frac{A}{2} = \frac{1}{2}\int_0^{\frac{\pi}{3}} \left((1+\cos\theta)^2 - \left(\frac{3}{4\cos\theta}\right)^2\right) d\theta$$

$$A = \frac{8\pi + 9\sqrt{3}}{16}$$

---

8) $y = 2x^2$, $y = 0$, $y = 5$ ile sınırlı bölgenin $x = 6$ eks. etr. dön. el. cismin hacmi?



dış ξ — sheel

$$V = \pi\int F^2(x)\,dx \quad , \quad V = 2\pi\int r\,h\,dx$$

$r = 6 - x$

$h = 5 - 2x^2$

$$V = 2\pi\int_{-5}^{5} (6-x)(5-2x^2)\,dx$$

$$= 2\pi\int_{-5}^{5} 30 - 12x^2 - 5x + 2x^3 = 2\pi\int_{-5}^{5} (2x^3 - 12x^2 - 5x + 30)\,dx$$

